THE CO-OPERATIVE UNIVERSITY OF KENYA

Lecture 01 — Foundations, Lifecycles, and Ecosystems

1. Core Definitions: What is Data Science?

Data Science is fundamentally the **science of data**. The entire field relies on two key pillars: **Data** and **Science**.

- **Data:** Raw, unorganized facts about something. Before it is processed, it has no meaning.
- **Science:** A step-by-step way of asking questions, running experiments, and finding the truth.

The DIKW Climb: How Data becomes Wisdom

Data doesn't stay raw forever. In your career, you will systematically move data up this 4-step ladder:

Data → Information → Knowledge → Wisdom

- **Data (Raw Clues):** Random numbers or text with no context. *Example: 39*
- **Information (The Context):** Giving the data a label so it makes sense. *Example: "The patient's body temperature is 39°C"*
- **Knowledge (The Learning):** Connecting that information to what you know about the world. *Example: "A temperature of 39°C means the patient has a high fever."*
- **Wisdom (The Action):** Using that knowledge to make a smart decision or improve a situation. *Example: "Prescribe fever-reducing medicine immediately."*

2. Defining "Big Data" & Data Mining

When data becomes massive, diverse, and fast-moving, we call it **Big Data**. To qualify as Big Data, it must exhibit five key characteristics:

The 5 Vs of Big Data

- **Volume:** Massive, overwhelming amounts of data. *Example: Billions of credit card transactions processed worldwide every single day.*
- **Variety:** Data coming in many different types. *Example: A mix of text messages, voice notes, CCTV video clips, and JPEG pictures.*
- **Velocity:** The immense speed at which new data is generated and changes. *Example: Thousands of live status updates and tweets flooding the internet every second.*

 0722499397 skarume@cuk.ak.ke

- **Veracity:** The truthfulness and cleanliness of the data. Raw data is often messy and full of errors. To improve data veracity, we do data cleaning. *Example: Removing duplicate records, fixing typos, or deleting broken lines from a user dataset.*
- **Value:** Data containing hidden, highly useful insights. We use Data Mining to extract this value. *Example: Spotting a hidden buying trend that helps a company increase its monthly revenue.*

Data Mining vs. Analysis vs. Analytics

- **Data Mining:** The overarching goal of discovering hidden, previously unknown patterns, anomalies, and correlations within massive datasets.
- **Data Analysis:** Looking backward at historical data to explain what happened and why. *Example: Looking at last month's sales spreadsheet to figure out why a business lost money.*
- **Data Analytics:** Looking forward using data, statistics, and machine learning to predict what will happen next. *Example: Analyzing historical ride patterns to predict which parts of the city will need more Uber drivers at 9:00 PM tomorrow.*

Key Techniques Used in Data Mining

- **Data Summarization:** Condensing raw data into manageable insights using descriptive statistics. This includes finding measures of central tendency (mean/average, median, mode) or dispersion (variance, standard deviation). *Example: Finding the average height of all students in the class rather than looking at 100 individual heights.*
- **Data Transformation:** Converting raw data into clean formats or structures so computer algorithms can read them efficiently. *Example: Changing text ratings like 'Good, Better, Best' into numbers like 1, 2, 3.*
- **Data Visualization:** Turning complex numbers into visual graphics to make underlying trends instantly visible to the human eye. *Example: Displaying monthly profits using a clean, colorful bar chart instead of a boring text table.*

3. The Evolution of Data Science

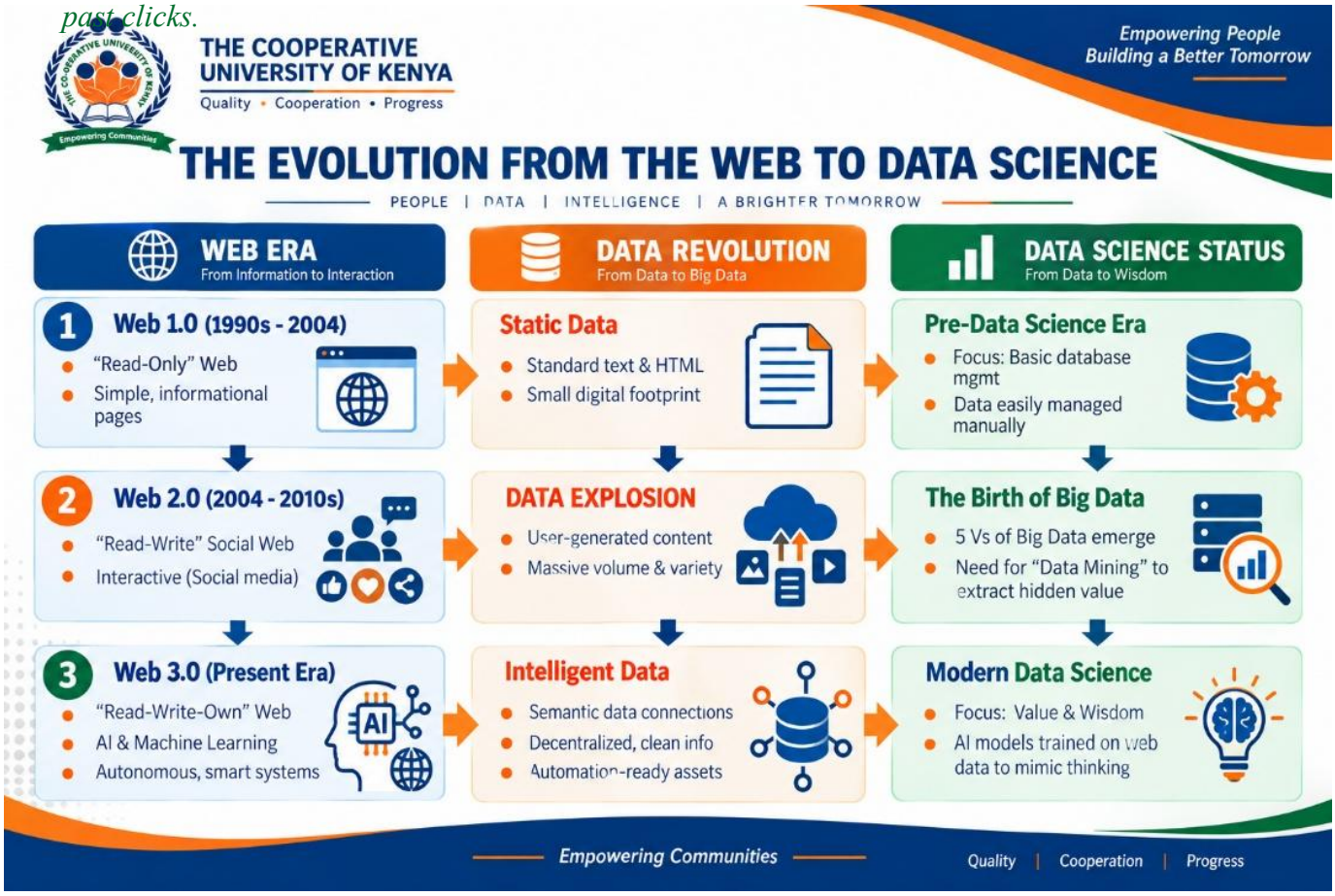
The field of data science emerged progressively from the data explosion on the internet, evolving hand-in-hand with the World Wide Web:

 0722499397 skarume@cuk.ak.ke

Web 1.0 (The Static Web): Purely informational and 'read-only.' Users could only read information but could not interact with it or upload anything. *Example: An old online encyclopedia where you could view static pages but could not comment or edit.*

Web 2.0 (The Transactional/Communicative Web): The 'read-write' web. Users actively create and input data into the web. This gave rise to Wikipedia, blogs, and social media tools, resulting in the data explosion that turned massive data into a highly valuable resource. *Example: Millions of users constantly uploading photos, writing posts, and leaving comments on Facebook or TikTok.*

Web 3.0 (The Intelligent Web): The modern 'smart' web. It actively utilizes that massive pool of data, advancing the fields of Artificial Intelligence (AI) and Machine Learning to make web systems autonomous, decentralized, and intelligent. *Example: Smart systems like ChatGPT that read billions of web pages to talk to you like a human, or Netflix suggesting a movie based on your past clicks.*



4. The 8 Steps of a Data Science Project

When you solve a real-world problem for a client, you will follow these eight systematic steps:

1. Problem Definition: Understanding the client's needs, identifying the business problem, and translating it into a clear data science objective. *Example: A car dealership wants to know exactly how much to sell a used car for so they can make a profit without driving away customers.*

2. Data Acquisition: Gathering raw data from various sources (such as databases, APIs, or web scraping) and organizing it for storage. *Example: Writing code to automatically download a list of thousands of used cars, their ages, and their selling prices from an online marketplace.*

3. Data Preparation: Cleaning the data, handling missing values, and formatting transformations. This step directly fixes the Veracity issue of Big Data. *Example: Finding rows where the car's mileage is missing or where someone accidentally typed the price as a letter instead of a number, and fixing it.*

4. Exploratory Data Analysis (EDA): Examining patterns, spotting anomalies, and gaining a deep understanding of which features or variables are relevant to the problem. *Example: Creating a quick plot to see if older cars generally have lower prices, verifying if the data follows standard real-world patterns.*

5. Data Modelling: Figuring out how variables relate to the problem. At this step, we apply statistical methods or machine learning algorithms to discover the mathematical model or formula that solves the problem and makes predictions.

📌 PROBLEM SCENARIO: CAR PRICE VALUATION TOOL

A car owner would like to understand the value of his second-hand car. How can data science help to create a car price valuation tool?

To solve this problem as a data scientist, you need to ask the following questions:

1. What data or features of a used car determine its selling price? (e.g., mileage covered, body color, body condition, car model, fuel type, etc.)

2. How can these features be related to the car price mathematically?

This is where you establish the Data Model formula:

Car Price = f (Mileage, Age, Brand, Fuel Type, Body Condition, ...)

Where f stand for function of

 0722499397 skarume@cuk.ak.ke

- **6. Model Evaluation:** Testing the model using metrics to check its accuracy and reliability before showing it to the client. *Example: Feeding the formula 50 old car profiles it hasn't seen before to check if its calculated prices are accurate or wildly wrong.*
- **7. Data Visualization & Communication:** Translating complex technical findings into clear charts, graphs, and stories to effectively communicate insights to the client. *Example: Presenting a simple, visual chart to the dealership manager showing exactly how car prices drop as mileage increases.*
- **8. Model Deployment & Maintenance:** Putting the final solution into live production so it runs automatically in the real world, while continuously monitoring it. *Example: Integrating your pricing formula into the dealership's smartphone app so salespeople can instantly type in a car's details and get a suggested price on the spot.*

5. What are Open-Source Tools?

Open-source tools are software applications whose source code is public and completely free for anyone to inspect, modify, use, and enhance.

The Economics of Open-Source Tools

Because open-source tools cost nothing in licensing fees, they are highly economical. They allow individual data scientists, startups, and massive enterprises to build advanced data systems without expensive software budgets.

- **Example:** *A student can download the exact same, world-class coding tools used by developers at Google or Meta without paying a single shilling.*

Categorization of Data Science Tools

- **Collaborative Tools:** Platforms that allow data teams to work together seamlessly, share code, track project changes, and document findings in a shared space. Examples: GitHub or Jupyter Notebooks.
- **Step-Aligned Tools:** Specialized software engineered to handle the specific technical phases of a project lifecycle.

 0722499397 skarume@cuk.ak.ke

Your Step-Aligned Toolkit & Examples

- **Tools for Data Acquisition (Step 2):** BeautifulSoup & Scrapy (Python libraries used to scrape text, prices, or files out of public web pages); Apache Kafka (captures massive streams of live, fast-moving data). *Example: Uber uses Apache Kafka to ingest millions of live GPS location updates from active drivers every single second.*
- **Tools for Data Preparation (Step 3):** Pandas (organizes data into clean tables of rows and columns); Weka (a point-and-click graphical tool to clean data visually without code). *Example: Loading a messy customer list into Pandas and writing one line of code to find and delete 500 duplicate accounts instantly.*
- **Tools for Exploratory Data Analysis - EDA (Step 4):** Jupyter Notebooks (a web application where you write code, hit run, and immediately see visual outputs underneath). *Example: Importing a university dataset into a Jupyter Notebook to instantly display the minimum, maximum, and average grades of a class.*
- **Tools for Data Modelling (Step 5 & 6):** Scikit-learn (for traditional machine learning like regressions); TensorFlow & PyTorch (for building artificial neural networks). *Example: Feeding 5,000 historical house prices into Scikit-learn to train a mathematical model that predicts future house values based on size.*
- **Tools for Data Visualization (Step 7):** Matplotlib & Seaborn (Python graphing libraries); Apache Superset (a platform for building interactive business dashboards). *Example: Using Seaborn to draw a quick scatter plot showing a direct visual line of how a car's value drops as its mileage increases.*
- **Tools for Model Deployment (Step 8):** Docker (packages your code, libraries, and settings inside a secure container box so it runs flawlessly on any server). *Example: Building a movie recommendation algorithm on a Windows laptop, then locking it in a Docker container so it runs perfectly on Netflix's Linux cloud servers.*
-

 0722499397 skarume@cuk.ak.ke